PLASMA PHYSICS & TOKAMAK CONFINEMENT — GRADUATE EXAM

Total Points: None | Questions: 5 | Date: April 17, 2026

Diagram-based questions on Plasma Physics & Tokamak Confinement

Question 1

10.0 points

Analyze the tokamak cross-section in the diagram below, which illustrates both toroidal and poloidal magnetic fields. Explain how these fields contribute to plasma confinement and identify any field lines that could lead to instabilities.

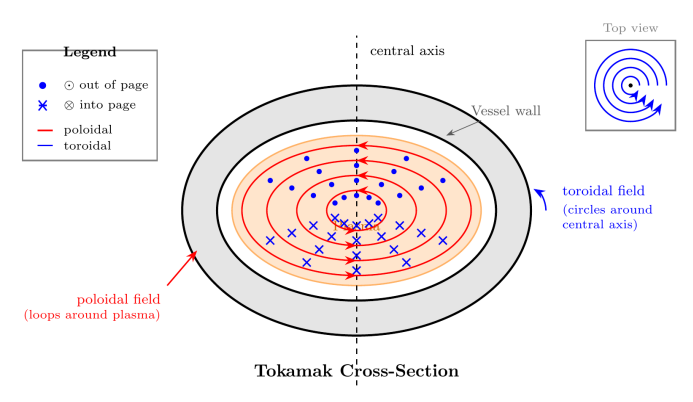

Fig. 1. A cross-section of a tokamak showing both toroidal and poloidal magnetic field lines. The toroidal field lines should be depicted as concentric circles around the central axis of the tokamak, while the poloidal field lines should be shown as loops that encircle the plasma vertically. The diagram should clearly label the toroidal and poloidal fields and highlight any field lines that could potentially lead to instabilities in plasma confinement. IMPORTANT: Values or sequences in the diagram don't match the question. Specifically: The diagram description requires toroidal field lines to be shown as concentric circles and poloidal field lines as vertical loops.. The diagram's depiction of the magnetic fields directly contradicts the requirements in the diagram description, swapping the visual representations for the toroidal and poloidal fields.. Regenerate using EXACTLY the values and sequences specified. IMPORTANT: Values or sequences in the diagram don't match the question. Specifically: MISMATCH 1: diagram shows toroidal field lines as vertical loops but question requires. The diagram incorrectly depicts the toroidal field lines as vertical loops and the poloidal field lines as concentric circles, which is the exact opposite of the required physical representation specified in the diagram description.. Regenerate using EXACTLY the values and sequences specified.

Question 2

10.0 points

Examine the Lawson criterion diagram below showing the triple product $nT\tau$ versus temperature. Explain the significance of the regions above and below the breakeven line for magnetic confinement fusion.

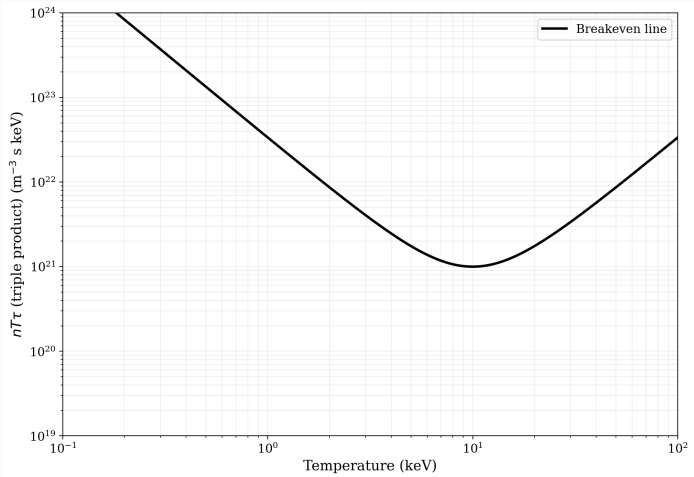

Fig. 2. Lawson criterion plot showing the triple product nTτ versus temperature with a breakeven line. The plot should include labeled axes for nTτ and temperature, with the breakeven line clearly marked. Regions above and below the breakeven line should be distinguishable to illustrate their significance for magnetic confinement fusion.

Question 3

10.0 points

Given the diagram below illustrating the magnetic field curvature and the grad-B drift of charged particles in a tokamak, describe how these factors impact particle confinement and drift in plasma.

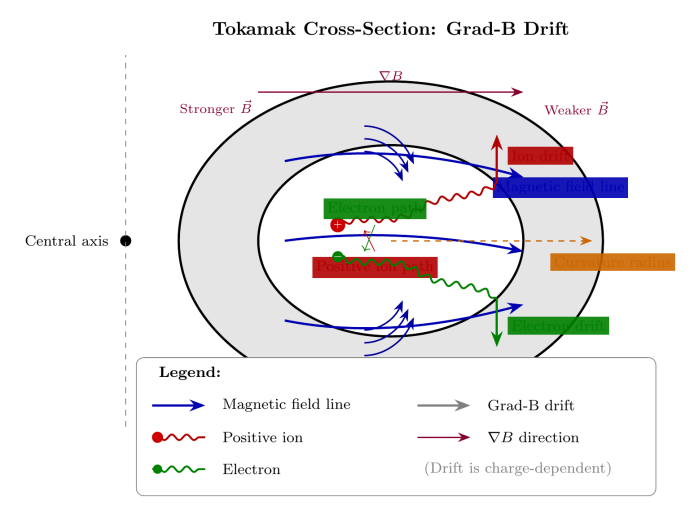

Fig. 3. Diagram illustrating magnetic field curvature and grad-B drift in a tokamak. Show a cross-section of a tokamak with magnetic field lines curving around the torus. Indicate the direction of the magnetic field with arrows. Include charged particles experiencing grad-B drift, with arrows showing their drift direction perpendicular to both the magnetic field and the gradient. Label the magnetic field lines, grad-B drift, and particle trajectories to highlight their interactions. IMPORTANT correction needed: The diagram is unreadable due to multiple labels overlapping each other, particularly in the legend, which makes it impossible to clearly understand the key.. Specifically: In the legend, the label "Grad-B drift" is written directly on top of "Magnetic field line". Similarly, the label "Drift direction (charge dependent)" is written on top of the "Positive ion" and "Electron" labels. This severe overlap makes the legend confusing and difficult to interpret.. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only. IMPORTANT: Values or sequences in the diagram don't match the question. Specifically: The label "Positive ion path" overlaps with "Weaker B". The label "Drift" (for the positive ion) overlaps with. The diagram contains multiple labels that are overlapping and unreadable, directly contradicting the explicit negative constraint in the prompt to avoid this specific issue.. Regenerate using EXACTLY the values and sequences specified.

Question 4

10.0 points

Analyze the q-profile (safety factor) and magnetic flux surface diagram for a tokamak shown in the diagram below. Explain how the safety factor q and magnetic flux surfaces are related to plasma stability.

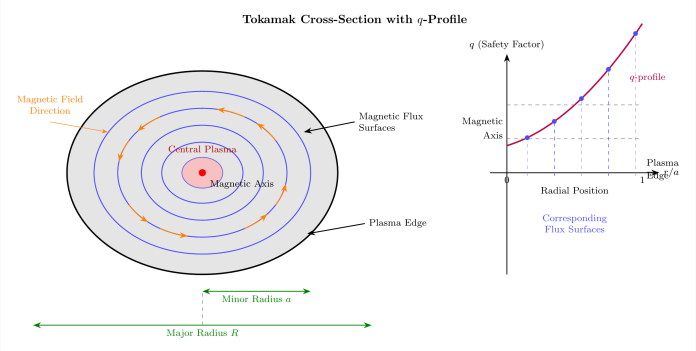

Fig. 4. Diagram showing a tokamak cross-section with magnetic flux surfaces and a q-profile. The tokamak is represented as a toroidal shape with concentric magnetic flux surfaces depicted as nested ovals. The q-profile is shown as a graph overlaying the magnetic flux surfaces, indicating how the safety factor q varies with the radial position within the tokamak. The diagram should include labels for the magnetic axis, plasma edge, and major/minor radii. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • stable The diagram will FAIL review if any of these appear. IMPORTANT: Values or sequences in the diagram don't match the question. Specifically: The diagram description requires "The q-profile is shown as a graph overlaying the magnetic flux surfaces", but the diagram only contains a. The diagram is missing the q-profile graph, which is explicitly required by the diagram description and is necessary for the student to answer the question.. Regenerate using EXACTLY the values and sequences specified.

Question 5 10.0 points

Examine the plot below that shows plasma instability growth rate against wave number. Discuss the implications of regions where the growth rate is positive, and propose methods to stabilize the plasma.

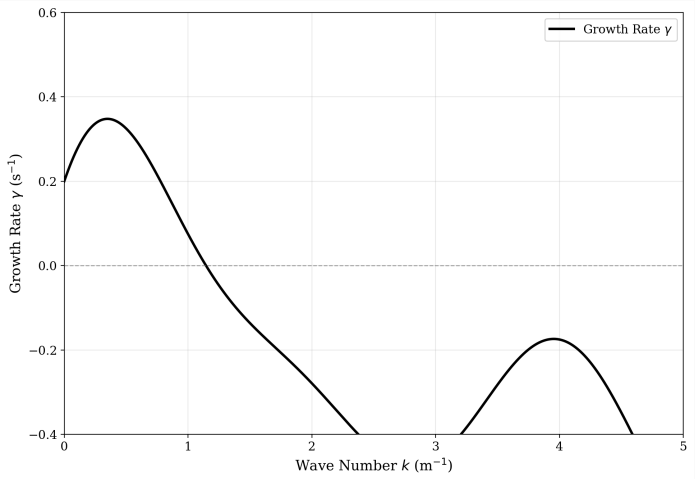

Fig. 5. A plot showing plasma instability growth rate as a function of wave number. The x-axis represents the wave number, and the y-axis represents the growth rate. The plot should highlight regions where the growth rate is positive, indicating instability, and regions where it is zero or negative, indicating stability. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • Stabilizing • stable • unstable The diagram will FAIL review if any of these appear.